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TwinShield-Fusion: Digital-Twin Residual Gating for Unseen Cyber-Physical Attacks on Electric Vehicle CAN Buses

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ABSTRACT An electric vehicle is a cyber-physical system: the commands that brake it, steer it and drive it are software messages on a Controller Area Network (CAN) bus that carries no sender identity and no authentication. An adversary who reaches that bus, through the diagnostic port, a compromised telematics unit or an aftermarket dongle, can forge any of those commands, and the vehicle obeys them. The protocol cannot be repaired without replacing every electronic control unit on the vehicle, so intrusion detection is the only defence deployable on vehicles already on the road. Published CAN detectors report accuracies above 99%, but they are trained and tested against a fixed catalogue of attacks: an attack outside that catalogue has no output class to fall into, so the detector reports it as normal traffic. This paper presents TwinShield-Fusion, which pairs a supervised CNN-BiLSTM classifier with a gated recurrent unit digital twin fitted on attack-free traffic alone. The twin predicts the next payload of each arbitration identifier, and the standardised per-identifier, per-byte prediction residual is thresholded at a calibrated false-positive rate, so the detector measures deviation from normal vehicle behaviour rather than similarity to a known attack. We evaluate on two real captures: GEM-CAN, 143,444 frames from a Polaris GEM e6 electric vehicle, and CICIoV2024, a capture from a 2019 Ford. With one attack class quarantined from every training stage on GEM-CAN, the three supervised baselines detect 0.84%, 0.00% and 25.21% of that class while TwinShield-Fusion detects 100.00% of it at a 2.26% false-positive rate. On CICIoV2024 the same protocol, applied once to each of three attack classes, detects 100.00% of each at a 0.37% false-positive rate. Every one of these figures is specific to the class named as quarantined in that run; none of them is a claim of universal zero-day detection. The boundary is measured rather than assumed, and we report it: on the same GEM-CAN capture the gate detects none of the steering-tampering class, whose forged payloads lie inside the range the twin already accepts, and on SynCAN, a synthetic benchmark composed entirely of attacks of that kind, it detects at most 7.07%. We further show that both public captures are far more redundant than their size suggests, CICIoV2024 containing only 3,588 distinct identifier-payload tuples across 1.4 million frames, so frame-level scores on them measure the benchmark as much as the detector.

INDEX TERMS Automotive cyber-physical security, controller area network, digital twin, electric vehicles, intrusion detection systems, residual anomaly detection, zero-day attacks.

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

An electric vehicle is not a mechanical machine with software bolted on. Its brake pedal is a sensor, its steering column is a motor, and its accelerator is a request. Between the driver and the wheels sit dozens of electronic control units (ECUs) that exchange several thousand messages per second over a Controller Area Network (CAN) bus standardised as ISO 11898-1 [1]. Electrified and drive-by-wire platforms deepen that dependence: regenerative braking blends friction and motor torque in software, electric power steering has no mechanical fallback, and the battery management system governs the current that moves the vehicle at all. Every one of those functions is actuated by a CAN frame.

CAN was designed in the 1980s for a closed network of trusted nodes. A frame carries an arbitration identifier, a data length code and up to eight payload bytes. It carries no sender address, no signature and no sequence number, and

every node receives every frame. Arbitration is won by the numerically lowest identifier, so a node that transmits identifier zero continuously starves every other transmitter on the bus. None of this is a defect of an implementation; it is the protocol, and it is the same protocol on vehicles sold today.

The consequences have been demonstrated repeatedly on production vehicles. Koscher et al. [2] showed that an attacker with bus access could disable the brakes and the engine of a modern car. Checkoway et al. [3] showed that the access itself can be obtained remotely, without physical contact. Miller and Valasek [4] drove an unaltered passenger vehicle off the road over a cellular link. Industry reporting has tracked the shift: an analysis of more than nine hundred automotive cybersecurity incidents found that 84.5% were carried out remotely [12]. An electric vehicle widens the attack surface further, because charging infrastructure, over-the-air update channels and telematics units are all network-facing and all eventually speak to the same internal bus.

The structural problem is that the bus cannot be fixed. Adding authentication to CAN means adding cryptographic material and processing budget to every ECU on the vehicle, which is neither retrofittable to the vehicles already on the road nor free on the ones being built. Intrusion detection is therefore not one option among several; for the installed fleet it is the only option. The question this paper asks is what such a detector must be able to do before it can be trusted on a vehicle.

B. RESEARCH CHALLENGES

Five properties of the problem make CAN intrusion detection harder than its reported accuracies suggest.

The attack catalogue is open-ended. A detector is trained on the attacks its authors could record. The adversary is under no obligation to use them. A supervised classifier has one output neuron per class it was taught, so an attack it has never seen cannot be assigned to a class that does not exist; in practice it is assigned to the majority class, which is normal traffic. The failure is silent, and it is the failure that matters operationally.

Benchmarks contain shortcuts. Volumetric attacks dominate the public captures, and a volumetric attack usually announces itself in the arbitration identifier. Section III-C shows that on GEM-CAN a lookup table over the identifier alone reaches 98.9% accuracy, so a deep model reported at 99% may have learned almost nothing a lookup table did not already know.

Normal traffic is repetitive and the classes are imbalanced. ECUs emit fixed status payloads at fixed intervals, so consecutive frames are nearly identical and a uniformly random split leaks near-duplicates between training and test. On GEM-CAN the ratio of normal to tampering frames is approximately 152:1, so an uninformative detector that always answers normal scores well on accuracy.

False alarms have a budget. A vehicle bus carries thousands of frames per second, so a false-positive rate that looks small in a table becomes an alarm stream no operator or downstream controller can triage. Any detector that buys sensitivity must report what that sensitivity costs.

Datasets are heterogeneous. Captures differ in vehicle, bus speed, identifier set, attack family and even in the fields they record; CICIoV2024, used in this paper, ships no timestamp at all. A method that is only shown to work on the capture it was designed around has not been shown to generalise.

C. RESEARCH GAP

Three gaps follow from those challenges and are visible in the literature summarised in Section II. First, evaluation is closed-set. The accuracies reported are recognition rates for attacks inside the catalogue, and they say nothing about attacks outside it, yet the headline claim is usually stated without that qualification. Second, supervised and unsupervised detectors are reported in separate papers, on separate splits, under separate metrics, so the comparison that a deployment actually needs, between a detector that names attacks and one that measures deviation, is never made on the same data. Third, validation is usually confined to one capture, and the few cross-dataset studies that exist report performance rather than the conditions under which a method transfers.

D. RESEARCH CONTRIBUTIONS

This paper makes the following contributions.

1) TwinShield-Fusion, a detector that combines a supervised CNN-BiLSTM classifier with a GRU digital twin of normal CAN behaviour. The standardised per-identifier, per-byte prediction residual is read through two views, a per-frame view and a per-ECU-stream view, and thresholded at a false-positive rate calibrated on normal traffic alone. The residual gate is the component we claim; the attention-weighted multi-ECU fusion is retained from the backbone and we show in Section V-G that it contributes no measurable gain in these experiments.

2) A leave-one-attack-out protocol with a four-stage quarantine. The held-out class is removed from the label space, from every training context window, from the digital twin's per-ECU history and from the gate's calibration set, so no component sees a single frame of it during fitting.

3) An arbitration-identifier bias audit that quantifies the shortcut in the benchmark itself, together with an identifier-free control variant of every experiment.

4) Evaluation on two real vehicle captures, GEM-CAN from a Polaris GEM e6 electric vehicle and CICIoV2024 from a 2019 Ford, under the same protocol, with a converter that maps a third-party capture onto the schema this pipeline reads.

5) An explicit, measured statement of the method's operating condition: the residual gate detects attacks that place implausible values on the bus, and does not detect attacks whose forged values lie inside the learnt-normal range. We report the case where our own method fails rather than omitting it.

6) The false-alarm cost of the second channel, expressed as a rate, as alarms per unit time on a bus of a given frame rate,

and as a detection-against-false-alarm curve over the whole threshold range, so that the operating point can be judged against a triage budget rather than against a single table entry.

7) A redundancy and leakage audit of both public captures, which finds that 87.4% of GEM-CAN frames and 99.9% of CICIoV2024 frames are byte-identical to another frame, and that the whole CICIoV2024 release contains 3,588 distinct identifier-payload tuples in 1.4 million frames. Frame-level scores on these benchmarks have to be read in that light, including ours.

II. RELATED WORK

Research on in-vehicle intrusion detection divides along two axes: what the detector learns from, and what it is evaluated against. We review the field along the first axis in Sections III-A to III-C and along the second in Section II-D, because the second is where this paper's contribution sits.

A. CAN BUS INTRUSION DETECTION

The earliest detectors used properties of the bus rather than of the payload. Muter and Asaj [14] applied entropy to the message stream, on the reasoning that injected traffic lowers the entropy of an otherwise highly regular bus. Lee et al. [13] built OTIDS around the remote-frame response offset, exploiting the fixed timing with which an ECU answers a request. Cho and Shin [16] went below the protocol entirely and fingerprinted ECUs from clock skew, so that a frame claiming an identifier its usual transmitter did not send can be recognised. These methods need no labelled attacks and are cheap, but each covers one narrow mechanism: entropy and timing detect volumetric injection well and a low-rate payload forgery on a legitimate identifier hardly at all.

Classical supervised learning followed, on handcrafted frame features. Bari et al. [6] compared support vector machines, decision trees and k-nearest neighbours on real vehicle captures, and Lokman et al. [8] surveyed the family. Accuracies are high and the models are interpretable, but the features are chosen for the attacks in hand and the detectors inherit the closed-set limit discussed in Section I-C.

B. LEARNING-BASED CAN TRAFFIC ANOMALY DETECTION

Deep models removed the feature-engineering step. Kang and Kang [5] applied a deep neural network to CAN payloads; Song et al. [9] used a deep convolutional network over sequences of frames; Seo et al. [7] trained a generative adversarial network, GIDS, to recognise injected traffic. Rai et al. [10] compared LSTM, GRU and BiLSTM detectors on CAN captures and reported strong sequence-learning performance, and Wang et al. [11] added data augmentation for connected autonomous vehicles. Lightweight and adaptive variants continue the line: Xu et al. [15] proposed a dynamically adaptive detection framework intended for the compute budget an in-vehicle unit can actually carry.

A second strand represents normal traffic and thresholds the deviation from it. Taylor et al. [18] trained a recurrent network to predict the next payload of an identifier and flagged large prediction errors. Shahriar et al. [17] built CANShield around multi-view reconstruction of the signal space. The closest antecedent to this work is CANet [19], which learns a joint model of CAN signals with per-identifier subnetworks and scores reconstruction error. These methods are label-free and therefore not bounded by an attack catalogue, which is the property this paper builds on.

C. DIGITAL TWIN-BASED INTRUSION DETECTION

A digital twin is a model kept in correspondence with a physical system [20]. In industrial control, twins have been used to detect intrusion by comparing the plant's observed behaviour against the behaviour the twin predicts, so that the detector is defined by a model of correct operation rather than by a library of faults. The automotive literature has largely used twins for prognostics, calibration and fleet analytics rather than for security, and where behavioural deviation is used for CAN security it is usually a reconstruction error rather than a twin maintained per ECU.

The distinction matters for this paper. A twin fitted on attack-free traffic defines normality per arbitration identifier, so its residual is defined for any frame whatsoever, including a frame belonging to an attack class that has never been labelled. That is the property a closed-set classifier cannot have, and it is why the residual rather than the classifier carries the zero-day result in Section VI.

D. CROSS-DATASET EVALUATION OF CAN INTRUSION DETECTION SYSTEMS

Public captures for CAN intrusion detection include Car-Hacking and OTIDS from HCRL [13], the ROAD dataset of Verma et al. [21], the synthetic SynCAN benchmark released with CANet [19], GEM-CAN from an autonomous electric vehicle [22], and CICIoV2024 from the Canadian Institute for Cybersecurity [23]. They differ in vehicle, bus speed, identifier set, attack family and recorded fields, and that heterogeneity is the obstacle to cross-dataset work: a model trained on one capture's identifier vocabulary cannot be applied directly to another's.

Most published detectors are validated on a single capture. Where a second dataset is used, it is normally used to repeat a performance number rather than to test a hypothesis about when the method works. Table I summarises the studies discussed above by method, by the dataset each was evaluated on, and by whether any attack class was held out of training. The pattern is uniform: high accuracy, one or two datasets, and no held-out class.

TABLE I
SUMMARY OF REPRESENTATIVE CAN INTRUSION DETECTION STUDIES

Study	Method	Dataset	Evaluation limitation
Kang and Kang [5]	Deep neural net	OCTANE (sim.)	Closed set
Song et al. [6]	Inception-ResNet	Car-Hacking	Closed set
Seo et al. [7]	GAN-augmented CNN	Car-Hacking	Closed set
Bari et al. [12]	SVM, tree, kNN	Car-Hacking	Closed set
Lee et al. [13]	Remote-frame offset	OTIDS	Known types only
Müter and Asaj	Message entropy	Own capture	Volumetric only

Study	Method	Dataset	Evaluation limitation
[14]			
Shahriar et al. [15]	Signal autoencoders	SynCAN, ROAD	No quarantine
Taylor et al. [16]	LSTM prediction error	Own capture	No baseline comparison
Hanselmann et al. [17]	CANet autoencoder	SynCAN, real	No baseline comparison
Cho and Shin [18]	Clock-skew fingerprint	Own capture	Masquerading ECU only
This work	Head + twin residual gate	GEM-CAN	Single vehicle capture

Three observations follow, and together they form the gap this paper addresses. The accuracies are recognition rates inside a fixed catalogue. The unsupervised entries are trained on attack-free traffic but reported separately from the supervised ones, so the comparison that a deployment needs is never made on one split. And validation rarely crosses a dataset boundary, so the conditions under which a method transfers are unknown. We place a supervised classifier and an attack-agnostic residual gate on the same data, under one quarantine applied at four stages, on two real vehicle captures, and report both under a metric that is defined irrespective of label space.

III. DATASETS AND PROBLEM FORMULATION

A. GEM-CAN DATASET

GEM-CAN [22] is a public capture recorded by Tavasoli et al. from a Polaris GEM e6, an autonomous electric vehicle, on a closed rural test track. The vehicle carries a 250 kbit/s CAN bus and the traffic was recorded through a PCAN-USB interface connected via a PACMod drive-by-wire interface. Two captures are supplied: one of nominal autonomous driving, and one during which attacks were physically injected onto the live bus while the vehicle was stationary in autonomous mode, for safety. Every frame carries a timestamp, an arbitration identifier, a data length code, eight payload bytes and an attack-type label.

The capture suits the question of this paper for three reasons. It is not a simulation, so the payload dynamics are real. It is an electric, drive-by-wire platform, so the commands that are attacked are the commands that actuate the vehicle. And it contains both a volumetric attack and two low-rate tampering attacks on legitimate identifiers, which lets one dataset exercise both the easy and the difficult detection regimes. Its composition is given in Table II.

TABLE II
COMPOSITION OF THE GEM-CAN CAPTURE

Class	Frames	Arbitration IDs used
Normal	101,039	13 distinct IDs
DoS	41,080	00000000 only
Brake tampering	663	18FF00F9 (shared with normal)
Steering tampering	662	18FF70F9 (shared with normal)
Total	143,444	—

The two attack families differ in a way that determines everything that follows. The denial-of-service attack transmits arbitration identifier 00000000. Because CAN arbitration is won by the lowest identifier, that frame takes priority over all legitimate traffic and the bus load rises from 20% to 100% during the flood. It is trivially separable: identifier 00000000 is sent by no legitimate ECU on this vehicle, so the identifier alone identifies the attack. The tampering attacks insert a small number of forged frames on 18FF00F9 and 18FF70F9, identifiers that carry real traffic throughout the capture, and the recorded effect is a steering-lock delay of about 20 seconds. Here the identifier is uninformative as a discriminator and detection must rest on the payload.

B. CICIoV2024 DATASET

CICIoV2024 [23] is a capture released by the Canadian Institute for Cybersecurity from a 2019 Ford. It contains benign traffic and five attack classes: a denial-of-service flood and spoofing of the gas, RPM, speed and steering-wheel signals. The published composition is 1,223,737 benign frames, 74,663 DoS frames, 9,991 gas-spoofing, 54,900 RPM-spoofing, 24,951 speed-spoofing and 19,977 steering-wheel-spoofing frames. The spoofing classes are the interesting ones for this paper, because each forges a signal on an identifier that already carries legitimate traffic, which is the regime in which the arbitration identifier carries no information.

Two properties of the release shape how it can be used, and both are stated here rather than left implicit. First, the schema is identifier, eight payload bytes and three label columns; it carries neither a timestamp nor a data length code. We therefore synthesise a timestamp from row order at a nominal one-millisecond interval and fix the data length code at eight. The consequence is that the inter-arrival feature is constant on this dataset and carries no information, so no timing-based claim is made on it. Second, the label space of this work has three attack slots and CICIoV2024 has five attack classes, so the four spoofing classes are grouped: steering-wheel spoofing occupies one slot on its own, and gas, RPM and speed spoofing are grouped as powertrain spoofing in the other. The assignment is recorded in machine-readable form by the converter and every result in Section VI-B is reported under the CICIoV2024 names.

C. DATASET CHARACTERISTICS AND ATTACK CATEGORIES

Table III sets the two captures side by side. They agree on the structural features that matter to the method and disagree on almost everything else, which is what makes the pair useful: GEM-CAN is an electric, drive-by-wire research platform with four classes and a severe identifier shortcut, while CICIoV2024 is a production passenger vehicle with six classes, no timestamp and a far larger benign population.

Before training anything we audited both captures for shortcuts, and found a severe one in GEM-CAN. A lookup table that guesses the majority class of each arbitration identifier, using no other information and no learning whatsoever, reaches 98.9% accuracy on the test split. It scores a per-class F1 of 1.00 on DoS and 0.00 on both tampering classes, giving a macro-F1 of 0.498. Two things

follow, and they shape the remainder of this paper. This is not a dataset with a meaningful accuracy headline, so macro-F1 and per-class figures are reported throughout. And the usefulness of any payload-modelling component has to be judged on the tampering classes, because DoS is already solved by a lookup table. We therefore retrain every model with the arbitration identifier removed from the input as well, which bounds how far the shortcut can be credited to a model rather than recognised as a property of the capture.

A second property of GEM-CAN is relevant to every figure that follows: the data are highly redundant. 87.4% of frames are byte-identical to at least one other frame, and 80.6% of test frames are exactly identical to a frame in the training split. This is intrinsic to CAN, where ECUs emit fixed status payloads at fixed intervals, but it makes any frame-level metric look better than the underlying difficulty warrants, and is another reason not to lead with accuracy.

TABLE III
CHARACTERISTICS OF THE TWO EVALUATION DATASETS

Property	GEM-CAN	CICIoV2024
Vehicle	Polaris GEM e6 (electric, drive-by-wire)	2019 Ford (production)
Setting	Closed rural test track	Road capture
Bus speed	250 kbit/s	not stated
Frames used	143,444	259,977
Classes	4	6 (grouped to 4)
Timestamp	recorded	absent; synthesised
Data length code	recorded	absent; fixed at 8
Volumetric attack	DoS on ID 00000000	DoS on one ID
Spoofing attacks	brake, steering	gas, RPM, speed, steering
ID-only accuracy	98.9%	100.0% (macro-F1 1.000)

Across both captures the attacks fall into two categories, and the distinction is more useful than the dataset-specific names. A volumetric attack injects frames at high rate, usually on an identifier that is either illegitimate or rarely used, and changes the statistics of the bus as a whole. A tampering or spoofing attack writes forged payloads onto an identifier that carries real traffic, at a rate low enough to leave the bus statistics intact, and changes only what the receiving ECU is told. Volumetric attacks are easy for any detector, including a lookup table. Spoofing attacks are the ones that require a model of what the payload should have been, and they are the attacks against which this paper measures itself.

A third benchmark is used in a supporting role and is named here so that the evaluation is not overstated. SynCAN [17], released with CANet, is synthetic rather than a vehicle capture: it provides signal values rather than payload bytes, and its attack types hold a signal at a constant valid value, drift it slowly, replay it or suppress it. It is used in Section V-C only, to test the operating condition of the residual gate against a family of attacks that the two real captures do not contain. Results from it are reported as results from a synthetic benchmark, never alongside the real-vehicle figures.

One property of both real captures bears directly on how their scores should be read. CAN traffic is highly repetitive by design, and these releases are more repetitive than their frame counts suggest. On GEM-CAN, 87.4% of frames are byte-identical to at least one other frame and 80.6% of test frames have an exact counterpart in the training split. On CICIoV2024 the effect is far stronger: the entire published release of 1,408,219 frames contains only 3,588 distinct identifier-and-payload tuples, and in the converted capture used here 99.9% of frames are duplicates of another frame with 99.8% of test frames having an exact counterpart in training. The attack classes are the most degenerate part of it, with 19,977 steering-spoofing frames drawn from three distinct tuples and 9,991 gas-spoofing frames from two.

The consequence is stated plainly because it applies to our own results as much as to anyone else's. A frame-level classifier on CICIoV2024 is close to a lookup over a few thousand tuples, so near-perfect frame-level scores on that capture are weak evidence of detection capability. We therefore treat GEM-CAN, which is less degenerate and which contains tampering attacks on identifiers that also carry legitimate traffic, as the load-bearing evaluation, and report CICIoV2024 as corroboration whose benchmark limitations are visible rather than hidden.

D. PROBLEM DEFINITION AND THREAT MODEL

Let a capture be a sequence of frames f_t = (tau_t, i_t, d_t, p_t), where tau_t is the timestamp, i_t the arbitration identifier, d_t the data length code and p_t the eight payload bytes. Each frame carries a label y_t from a finite set that contains Normal and a number of attack classes. The detector observes a context window of the W frames up to and including f_t and must emit a decision for f_t.

We ask two questions of that detector. The closed-set question is the usual one: given every class at training time, predict y_t. The open-set question is the one this paper is built around: one attack class is designated unknown and removed from every fitting stage, and the detector, which has no output index for it, must still answer whether frame f_t should raise an alarm. Because one model then has a label the others lack, accuracy and F1 are not comparable across models in that setting; the alarm rate is, because it is defined in any label space.

The adversary is assumed to have already obtained write access to the bus, whether through a physical diagnostic port, a compromised telematics unit or an aftermarket device, which is the access level demonstrated in [2], [3] and [4]. The adversary can jam the bus, and can transmit arbitrary payloads on identifiers that legitimate ECUs already use. The detector is passive: it observes the bus and cannot authenticate senders, because the protocol provides no mechanism for it. Attacks that do not modify the bus, such as passive eavesdropping, are outside this threat model, as is an attack arriving on an identifier the twin has never observed, which is noted as a limitation in Section VII-D.

IV. PROPOSED TWINSHIELD-FUSION FRAMEWORK

A. SYSTEM ARCHITECTURE

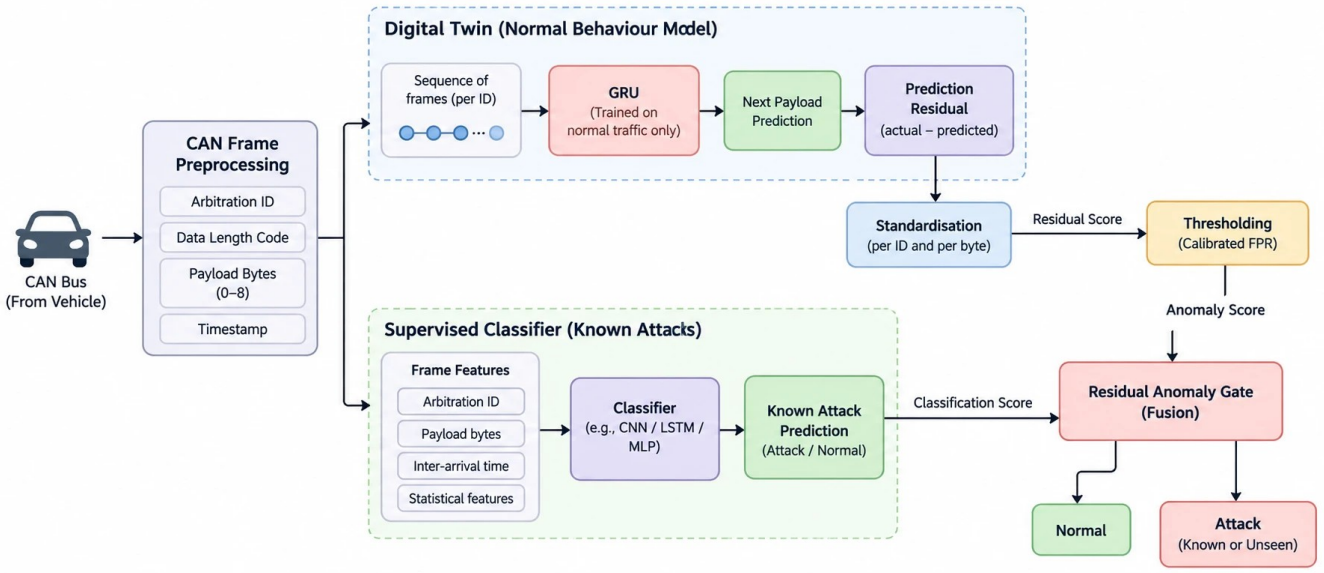


FIGURE 1. TwinShield-Fusion. Raw CAN traffic is preprocessed once and read by two branches. The digital twin predicts the next payload of each arbitration identifier from normal traffic alone, and its prediction residual is standardised per identifier and per byte and thresholded at a calibrated false-positive rate (Sections V-C and V-D). The supervised classifier, a CNN-BiLSTM in this work, names the attacks it was trained on (Section IV-E). The residual anomaly gate combines the two scores, which is what lets an attack with no label of its own still raise an alarm.

Fig. 1 shows the architecture. Raw CAN traffic is preprocessed once and read by two branches that answer different questions. The supervised branch builds a 32-frame context window and classifies the current frame with a CNN-BiLSTM enhanced by an attention-weighted fusion of per-ECU residual streams; it answers which known attack this is. The digital-twin branch predicts the next payload of each arbitration identifier from normal traffic alone and standardises the prediction residual per identifier and per byte; it answers how far this frame is from what the vehicle normally does. A gate thresholds the standardised residual at a false-positive rate calibrated on normal frames, and an alarm is raised if either branch fires.

The separation is the point of the design. The supervised branch is bounded by its label set and cannot generalise beyond the classes it was fitted on. The twin branch is fitted without any label at all, so its residual is defined for every frame, including one belonging to an attack class nobody has named. The detector can therefore say that something is wrong even when it cannot say what is wrong, which is the behaviour an operational system needs and a closed-set classifier cannot provide.

B. CAN TRAFFIC PREPROCESSING

Each frame is represented by its eight payload bytes scaled to [0, 1], its data length code, the inter-arrival gap to the previous frame on the same identifier, and an index for the arbitration identifier itself. A single frame is not enough to distinguish a forged command from a legitimate one on the same identifier, so the supervised branch reads a context window of 32 frames ending at the frame under test, in bus order. The unit of classification is therefore the window, and the contents of the window also dictate what must be excluded under the quarantine protocol of Section V-C.

On CICIoV2024 the inter-arrival gap is constant, because the release carries no timestamp and one is synthesised from row order. The feature is retained so that the two datasets pass through identical code, but it contributes nothing there, and the comparison in Section VI-C is read with that in mind.

C. DIGITAL TWIN-BASED TRAFFIC PREDICTION

Let p_t in [0, 1]^8 be the scaled payload of frame t and i_t its arbitration identifier. The twin is a gated recurrent unit [24] with 96 hidden units and a 16-dimensional embedding of the identifier, which lets one network serve every ECU on the bus while still conditioning on which ECU is speaking. It reads the eight most recent payloads observed on identifier i_t and predicts the next one. Training minimises mean squared error against the true next payload, over normal frames in the training split only.

The quantity the rest of the method uses is the residual

$$r_t = | p_t - \hat{p}_t | \quad (1)$$

taken element-wise, so r_t is an eight-dimensional vector with one component per payload byte. On GEM-CAN the twin reaches a mean absolute error of 0.041 in the units it optimises, or 10.52 on the 0-255 byte scale, measured on held-out normal frames. That is not a small error, and it does not need to be: detection rests not on a residual of zero but on a residual that is reliably different between normal and tampered traffic on the same identifier. CAN payloads multiplex independent signals, and some bytes are rolling counters or noisy sensor readings that are not predictable in principle.

D. RESIDUAL GENERATION AND ANOMALY DETECTION

The gate is the component that does not depend on known attacks. It reads the same residual through two views. The frame view asks how far this one frame departs from what its own identifier normally looks like, byte by byte:

$$z_frame(t) = max_b \, [\, r_\{t,b\} - \mu_\{i,b\} \,] \, / \, \sigma_\{i,b\} \quad (2)$$

where i is the identifier of the frame and b indexes the eight payload bytes. This is the sharp channel: a forged payload that moves one byte far outside its usual range shows up immediately and is not averaged away by surrounding traffic. The ECU view asks how far a whole identifier's stream has drifted across the context window:

$$z_ecu(t) = max_\{s \text{ present}\} \, [\, \bar{v}_s(t) - \mu_s \,] \, / \, \sigma_s \quad (3)$$

where $\bar{v}_s(t)$ is the mean residual of stream s inside the window. This is the sustained channel: it survives a single quiet frame and reflects the multi-ECU structure the fusion branch is built on. An alarm is raised when the larger of the two crosses a threshold:

$$alarm(t) = 1 \, [\, max(z_frame, z_ecu) > z^* \,] \quad (4)$$

Per-identifier, per-byte standardisation is what makes a single bus-wide threshold usable. Some ECUs and some bytes sit at a high baseline residual because they are intrinsically hard to predict; judging each byte of each identifier against its own normal behaviour prevents those from firing constantly. The means and standard deviations are estimated on normal frames of the training split, and z* is the (1 - alpha) quantile of the score over normal frames of the validation split, with alpha the target false-positive rate. No attack frame of any class takes part in fitting or calibrating the gate, and that is the whole basis for calling it attack-agnostic: there is no signature in it to memorise, only a model of normal.

E. INTRUSION CLASSIFICATION AND DECISION MECHANISM

The supervised branch is a convolutional network followed by a bidirectional long short-term memory network [25]. Three one-dimensional convolutional layers with 64 filters, kernels of width five and then three, read the context window and extract local byte patterns that are invariant to their position in the window, which is what a burst of injected frames looks like. The BiLSTM, with 96 hidden units in each direction, then reads the convolutional feature sequence both ways, so a frame is judged against traffic that follows it as well as traffic that precedes it.

Each arbitration identifier is treated as a separate ECU stream. Within a context window, stream s is summarised by its mean residual, giving a tensor over streams, and streams absent from the window are masked out. Additive attention [26] weights the streams,

$$e_s = v^T \tanh(W \, h_s + b), \quad \alpha = softmax(e) \quad (5)$$

and the fused vector sum_s alpha_s h_s is concatenated with the CNN-BiLSTM embedding before a softmax head over the trained classes. This branch names an attack and is bounded by its label set in exactly the way described in Section I-B; we treat it as the recognition channel and make no claim that it generalises beyond the classes it was fitted on.

The attention-weighted fusion is described here for completeness because it is part of the implementation, but it is not claimed as a source of the results. The ablation of Section V-G measures its contribution directly and finds it to be nil in both input variants, 0.9979 to 0.9979 with the identifier and 0.9979 to 0.9979 without it, and the decomposition of Section V-A shows that it contributes none of the quarantined-class detection either. The component this paper claims is the residual gate of Section III-D; readers reproducing the method should expect the residual, not the attention layer, to carry the result.

The two branches are joined by disjunction. A frame raises an alarm if the supervised head assigns it any class other than Normal, or if the gate fires. The disjunction is deliberate: the classifier is precise on what it knows and blind to the rest, the gate is broad and noisier, and a detector that must not miss an unknown attack should take the union rather than the intersection. The cost of that choice is quantified in Section VI-F.

F. MODEL EXPLAINABILITY

An alarm that cannot be explained cannot be triaged. The framework exposes two complementary explanations. The gate reports, for every alarm, which arbitration identifier and which payload byte carried the largest standardised residual, and which of the two views fired; that localises an alarm to an ECU and a signal without any additional model. For the supervised branch we compute Shapley additive explanation values [27] over its inputs, which attribute a decision across payload bytes, identifier and timing features.

Both explanations are partial in the same way, and it is worth stating. They identify where the evidence is, not why the twin finds a given payload surprising. A full account of the latter would require decoding the payload into physical signals, which needs the vehicle's signal database and is outside the scope of a detector that must work without one.

V. EXPERIMENTAL SETUP

A. EXPERIMENTAL ENVIRONMENT

All models are implemented in PyTorch and trained and evaluated on CPU. The main results use a fixed random seed of 42, and the leave-one-attack-out experiment on CICIoV2024 is additionally repeated with seeds 1, 2 and 3 so that the figures in Section V-B carry an uncertainty rather than resting on one draw. No GPU is used at any stage, which is deliberate: the runtime figures in Section VI-I are then measured on hardware comparable to what a vehicle or a workshop diagnostic tool could plausibly carry, rather than on an accelerator that no production ECU contains.

B. DATA PREPROCESSING AND FEATURE ALIGNMENT

Both captures are read into one schema: timestamp, arbitration identifier, data length code, eight payload bytes and a class label. GEM-CAN is used as published. CICIoV2024 is mapped onto the same schema by a converter that synthesises the timestamp from row order, fixes the data length code at eight, assigns each source class to a label slot as described in Section III-B, and records the assignment in a machine-readable file so that no figure drawn from the converted data can silently inherit the other dataset's class names. The converter verifies its own output by loading it back through the project's loader and refusing to proceed if any slot is empty.

From that common schema the feature matrix is identical for both datasets: eight payload bytes scaled to [0, 1], the data length code, the inter-arrival gap and an identifier index, assembled into 32-frame context windows. The identifier vocabulary is built per dataset, since the two vehicles share no identifiers.

C. TRAINING, VALIDATION, AND TESTING PROTOCOL

Consecutive CAN frames are almost identical, so a uniformly random split leaks near-duplicate frames between training and test and inflates every metric. Frames are therefore cut into contiguous blocks of 256 and whole blocks are assigned to training, validation and test in a 70/15/15 ratio, stratified so that every class is represented in each split. No frame's context window crosses a split boundary. This does not remove the correlation between adjacent blocks, and that residual limitation is reported in Section VII-D rather than hidden.

The leave-one-attack-out protocol is the central experimental instrument. One attack class is designated unknown and removed at four stages: the detector head is built with one fewer output so there is no index for it; every training and validation frame whose context window contains a frame of that class is dropped; the digital twin is refitted from scratch on normal frames whose per-ECU history is free of it; and the gate's means, standard deviations and threshold are taken from normal frames only. The test split is left untouched and still contains every frame of the held-out class.

D. BASELINE MODELS

Three supervised baselines are trained on the same splits, with the same class weights and the same evaluation code: a one-dimensional CNN, a BiLSTM, and a CNN-BiLSTM, the last of which is also the backbone of the proposed model. Two intermediate variants are reported for the ablation of Section VI-G: the backbone with a scalar twin output appended, and the backbone with the full per-byte residual vector appended. Every model is additionally re-run with the arbitration identifier removed from the input, because the audit of Section III shows the identifier alone is nearly sufficient on GEM-CAN.

All detectors use a 32-frame context window, a batch size of 512, the Adam optimiser at a learning rate of 10^-3, and class-weighted cross-entropy to compensate for the imbalance between normal and tampering traffic. The digital twin is trained for five epochs and the detectors for six, and the weights kept are those with the best validation macro-F1, which are then evaluated once on the test split.

E. EVALUATION METRICS

For the closed-set setting we report macro-F1, which weights every class equally and therefore does not let the dominant normal class conceal failure on the tampering classes, together with per-class F1 where the detail matters.

For the open-set setting macro-F1 across models is meaningless, because the proposed model has a label the baselines lack. We therefore report the alarm rate: the fraction of frames of a given true class on which the detector raised an alarm of any kind. On the held-out attack class this is the detection rate, and on the normal class it is the false-positive rate. Both are defined identically for every model regardless of label space, which is the property that makes the comparison fair. We additionally report the known-class macro-F1 of each model on the classes it did retain, so that a model cannot buy detection by degrading into an always-alarm detector without that being visible.

VI. RESULTS AND ANALYSIS

A. PERFORMANCE ON THE GEM-CAN TEST SET

Table IV gives closed-set macro-F1 for every model on the shared held-out test split, with and without the arbitration identifier as an input. With the identifier included, the four architectures span 0.0031 macro-F1 and all five rows span 0.0041. Neither number is evidence that one architecture is better than another; both are evidence that the benchmark has stopped measuring architecture. A lookup table on the identifier alone reaches 98.9% accuracy on this capture, so there is very little label information left for a sequence model to add.

TABLE IV
CLOSED-SET MACRO-F1 ON THE SHARED GEM-CAN TEST SPLIT

Model	CAN ID included	CAN ID excluded
CNN	0.9968	0.9914
BiLSTM	0.9948	0.9895
CNN-BiLSTM	0.9979	0.9914
GRU Digital Twin + CNN-BiLSTM	0.9989	0.9979
TwinShield-Fusion	**0.9979**	**0.9979**

Removing the identifier is more informative. There the digital-twin residual lifts the CNN-BiLSTM backbone from 0.9914 to 0.9979 macro-F1. That gain should be read against the support behind it: macro-F1 is weighted heavily by the two tampering classes, which contain 119 and 118 test frames, so a single frame changing class moves a tampering F1 by about 0.008 and macro-F1 by about 0.002. A difference of 0.0065 is therefore of the order of three frames. We report it as a consistent direction, not as a precisely resolved quantity.

The result that separates the models is the open-set one. Table VI reports the leave-one-attack-out experiment with brake tampering quarantined from every training stage. The

three supervised baselines raise an alarm on 0.84%, 0.00% and 25.21% of the 119 held-out frames; TwinShield-Fusion raises one on all 119. The known-class macro-F1 column confirms that the proposed model has not bought that detection by degrading on the classes it kept.

TABLE V
ABLATION ON THE SAME CNN-BILSTM BACKBONE, GEM-CAN (MACRO-F1)

Configuration	CAN ID incl.	CAN ID excl.
CNN-BiLSTM	0.9979	0.9914
+ digital twin (scalar)	0.9958	0.9965
+ residual vector	0.9989	0.9979
+ adaptive fusion	0.9979	0.9979

TABLE VI
LEAVE-ONE-ATTACK-OUT ON GEM-CAN: BRAKE TAMPERING QUARANTINED FROM EVERY TRAINING STAGE

Model	Detected	False alarms	Known macro-F1
CNN	0.84%	0.03%	0.6667
BiLSTM	0.00%	0.08%	0.7607
CNN-BiLSTM	25.21%	0.31%	0.7787
TwinShield-Fusion	**100.00%**	**2.26%**	**0.7829**

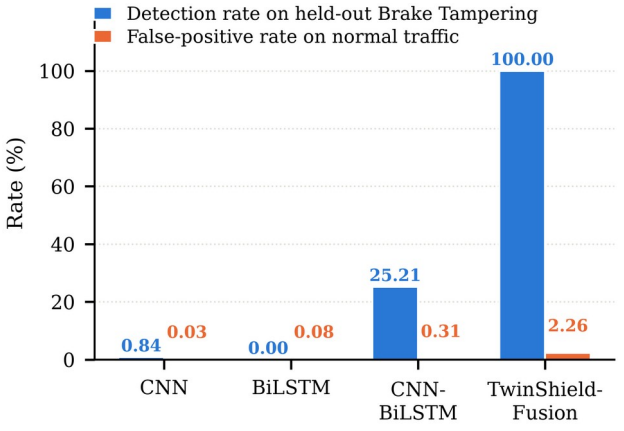


FIGURE 2. Detection of a quarantined attack class on GEM-CAN. Brake tampering was removed from the label space, from every training context window, from the digital twin's per-ECU history and from the gate's calibration set. Detection rate is measured on its 119 held-out test frames; the false-positive rate is measured on 15,063 normal test frames.

The credit belongs to one component and the decomposition is worth stating. TwinShield-Fusion's supervised head contributes none of that detection, because it has no label to assign; the anomaly gate contributes all of it. The residual is the generaliser and the classifier is the namer. The 25.21% achieved by CNN-BiLSTM is not recognition either: that model also has no output for brake tampering, and what it does is place some of those frames in steering tampering, the other tampering class it was trained on, which shares an identifier family and a timing signature. The alarm is raised by misattribution. The two weaker baselines cannot even do that.

B. PERFORMANCE ON THE CICIoV2024 TEST SET

The second capture asks whether the result survives a change of vehicle, of attack family and of recording conditions. Table VII gives closed-set macro-F1. With the arbitration identifier available every model scores 1.0000, and that is not a result about models: on this capture the identifier alone is a perfect classifier, because not one of the attack identifiers appears anywhere in the 1,223,737 benign frames of the release. The informative column is therefore the one with the identifier removed, and there the digital twin lifts the CNN-BiLSTM backbone from 0.9992 to a perfect 1.0000, with per-class F1 of 1.0000 on every class. The twin also fits this vehicle far more closely than the GEM e6, reaching a mean absolute error of 1.88 bytes on 26,880 held-out normal frames against 10.52 on GEM-CAN.

TABLE VII
CLOSED-SET MACRO-F1 ON THE CICIOV2024 TEST SPLIT

Model	CAN ID included	CAN ID excluded
CNN	1.0000	0.9989
BiLSTM	1.0000	0.9985
CNN-BiLSTM	1.0000	0.9992
GRU Digital Twin + CNN-BiLSTM	**1.0000**	**1.0000**
TwinShield-Fusion	**1.0000**	**0.9999**

Table VIII repeats the leave-one-attack-out protocol, once for each of the three attack slots. TwinShield-Fusion raises an alarm on 100.00% of the held-out frames in all three runs, at a false-positive rate of 0.37% on normal traffic, while the three supervised baselines reach 0.00%, 35.71% and 0.16% at best. Known-class macro-F1 remains 1.0000 in every run, so the detection is not bought by degrading on the classes that were retained. The margin is wide rather than marginal: against a gate threshold of $z = 4.18$, the mean standardised residual of the held-out class is 109 for DoS, 104 for powertrain spoofing and 152 for steering spoofing, and the per-frame view carries the whole of each decision.

TABLE VIII
LEAVE-ONE-ATTACK-OUT ON CICIOV2024: EACH ATTACK SLOT QUARANTINED FROM EVERY TRAINING STAGE IN TURN

Quarantined class	TwinShield-Fusion	Best baseline	False alarms
DoS	**100.00%**	**0.00%**	**0.37%**
Powertrain spoofing	**100.00%**	**35.71%**	**0.37%**
Steering spoofing	**100.00%**	**0.16%**	**0.37%**

Two qualifications belong with that table. The first is that the separation is complete over the whole threshold range, not only at the operating point: the area under the detection-against-false-alarm curve is 1.000 for all three classes, and 100% detection is still reached at the smallest false-alarm rate the sweep resolves. Fig. 6 plots it. That is a strong result, and it is also a warning sign.

The second is the redundancy documented in Section II-C. CICIoV2024 contains 3,588 distinct identifier-and-payload tuples in 1,408,219 frames, its attack classes are drawn from as few as two or three distinct tuples, and no attack identifier appears anywhere in the benign capture. A detector facing that capture is close to facing a lookup problem, and a perfect score on it should be read accordingly. We report these numbers because they are the honest context for the table above: the CICIoV2024 result confirms that the method transfers to a second vehicle and a second attack family, and it does not, on its own, establish detection capability at the level the raw figures suggest. The GEM-CAN experiment,

where the tampering attacks ride on identifiers that also carry legitimate traffic, is the harder and more informative of the two.

To put an uncertainty on those figures rather than relying on a single seeded run, the steering-spoofing quarantine was repeated with four random seeds. Detection is perfectly stable: 100.00% on every seed, with a standard deviation of zero, as are the known-class macro-F1 of 1.0000 and the area under the curve of 1.000. The false-alarm rate is not stable. Across the four seeds it takes the values 0.37%, 0.35%, 0.55% and 0.02%, a mean of 0.32% with a standard deviation of 0.22, so the operating point moves by more than an order of magnitude while the detection rate does not move at all.

That asymmetry is informative and we report it rather than quoting the best seed. It follows from how the threshold is set: the gate's quantile is read from whichever normal validation frames the split assigns, and on a capture whose payload vocabulary is as small as this one a different draw of validation blocks shifts the tail of the normal score distribution appreciably. The detection rate is insensitive to that shift because, as Fig. 6 shows, the attack and normal score distributions do not overlap at all here. On a harder capture the same sensitivity would move the detection rate as well, which is the reason the false-alarm figures in this paper should be read as an operating region rather than as a single number.

C. COMPARATIVE ANALYSIS ACROSS DATASETS

Table IX places the quarantined-class result from both vehicles side by side. Across two real captures and four held-out classes, TwinShield-Fusion raises an alarm on every held-out frame, while the best supervised baseline on any of them reaches 35.71% and is usually at or near zero. The false-alarm cost differs by nearly an order of magnitude between the captures, 2.26% on GEM-CAN against 0.37% on CICIoV2024, which tracks how well the twin fits each vehicle: a mean absolute error of 10.52 bytes on the GEM e6 against 1.88 on the Ford.

TABLE IX
CROSS-DATASET SUMMARY OF THE QUARANTINED-CLASS RESULT

Dataset / vehicle	Quarantined class	Detected	Best baseline
GEM-CAN, Polaris GEM e6	brake tampering	100.00%	25.21%
CICIoV2024, 2019 Ford	DoS	100.00%	0.00%
CICIoV2024, 2019 Ford	powertrain spoofing	100.00%	35.71%
CICIoV2024, 2019 Ford	steering spoofing	100.00%	0.16%

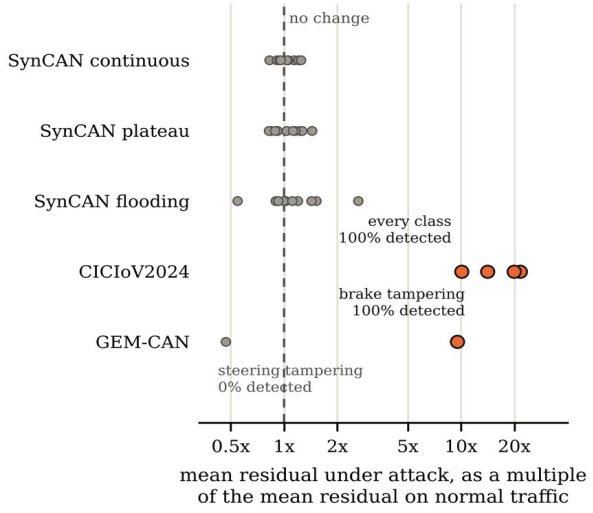


FIGURE 3. Why the gate transfers or does not. Each marker is one arbitration identifier under one attack class, placed at the ratio of its mean digital-twin residual under attack to its mean residual on normal traffic. Only an attack that drives the residual far outside its normal range is separable by a gate calibrated at a 1% false-alarm rate.

A third capture supports the same reading. SynCAN [19], the synthetic benchmark released with CANet, consists entirely of signal-space manipulations that hold a signal at a constant valid value, drift it slowly, replay it or suppress it. Converted to this schema and run under the identical protocol, the gate detects at most 7.07% of a quarantined class there, and the median ratio of attack residual to normal residual across thirty identifier-and-class pairs is 1.04. A control in which each signal is packed into one byte rather than two makes the twin five times more accurate and leaves detection unchanged at 1.56%, which rules out the quantisation as the cause. The pattern across all three captures is the same, and it is a property of the attack rather than of the dataset.

D. COMPARISON WITH BASELINE MODELS

Fig. 4 places the same four models under the two protocols. Under the closed-set protocol they are indistinguishable, separated by 0.0031 macro-F1. Under the leave-one-attack-out protocol on GEM-CAN they are separated by more than 74 percentage points. The comparison is the central argument of this paper: a benchmark on which four architectures agree to three decimal places is not measuring the property that matters, and changing the question rather than the model is what makes the difference visible.

The cost of the second channel is the false-positive cost. The always-on alarm detects all attacks, so a detection rate without false positive is not a result. On normal test traffic, ours is 2.26%, while the baselines range from 0.03% to 0.31%, or about 340 false alarms per 15,063 normal frames. This is the cost of the additional channel and it is not insignificant. The target rate at calibration is 1%, but the realised rate is higher due to the fact that the false alarms of the supervised head are also counted as false alarms of the gate, and due to the fact that the normal traffic used for validation is not the same as the normal traffic used for test.

2.26% is 22.6 false alarms per thousand normal frames. The unit that is operationally meaningful is alarms per unit time, which on a bus with f frames per second is 0.0226f. At any frame rate a production powertrain bus maintains, there are more alarms than an operator or a downstream controller could triage, so we present the gate at this operating point as a research instrument, not as a deployable alarm. The threshold is the knob available, and the trade-off a deployment must make is the one that the curve represents, as it moves down the curve by lowering the target false-positive rate. We have not determined what the tolerable alarm rate is on it, as it is dependent on a per-vehicle frame rate and a per-fleet triage budget that this capture does not provide. The two experiments are compared in Fig. 3 and the argument of this paper is summarised. The four models are not distinguishable on the closed set. On the quarantined class they are more than 74 percentage points apart. A closed set score is a score that only considers attacks that have already been classified.

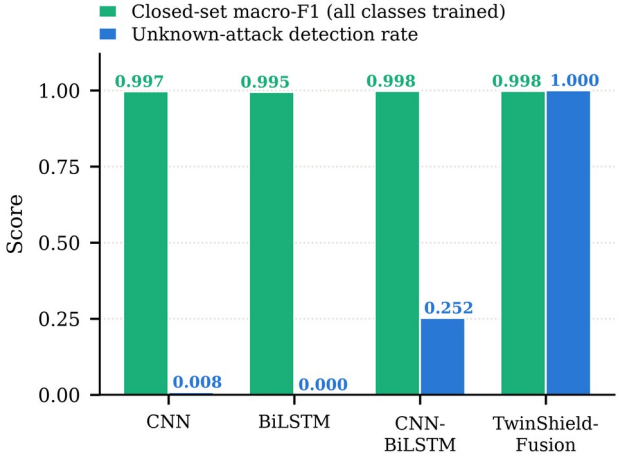


FIGURE 4. The same four models under two protocols on GEM-CAN. Closed-set macro-F1 separates them by 0.0031; detection of a quarantined attack class separates them by more than 74 percentage points.

The ordering of the four models is preserved on CICIoV2024, which matters more than the absolute figures: whatever the capture, the supervised baselines lose an unseen class and the residual gate is what recovers it, to the extent that the attack is recoverable at all.

E. ATTACK-WISE DETECTION PERFORMANCE

Aggregate numbers conceal the mechanism, so we report the residual by attack class and by arbitration identifier. The gate detects an attack if, and only if, that attack sends payload bytes the twin does not expect for that identifier. This is a real condition, not a formality, and Fig. 5 shows a case where it is satisfied and a case where it is not, on the same capture and the same model.

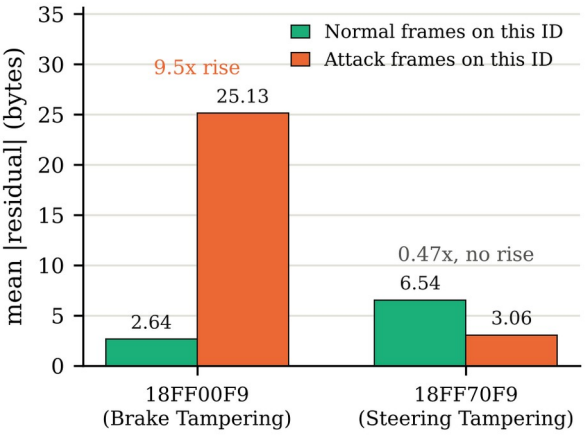


FIGURE 5. Mean digital-twin residual on the two GEM-CAN tampering identifiers. Brake tampering raises the residual by a factor of 9.5 and is detected. Steering tampering leaves it below the normal level and is not, because the forged payloads fall inside the range the twin already accepts for that identifier.

Under brake tampering the mean residual on identifier 18FF00F9 is 25.13 bytes, 9.5 times the 2.64 bytes measured on normal frames of that identifier, and the twin localises the attack precisely: bytes DATA_4 and DATA_5 carry almost all of the increase. On 18FF70F9 the mean residual under steering tampering is 3.06 bytes, below the 6.54 bytes of normal traffic on the same identifier. The forged steering payloads lie inside the range the twin already accepts, so the gate cannot detect steering tampering. That is not an experiment that failed; it is a measurement. A score above learnt-normal is a positive deviation, and a positive deviation is what the gate looks for.

This restriction is worth stating exactly, because it is the one most easily overstated. Detection does not require knowing what the attack is, but it does require the attack to be anomalous in the payload. Those are different conditions, and conflating them would overstate the method.

F. FALSE POSITIVE AND FALSE NEGATIVE ANALYSIS

The cost of the second channel is its false-positive rate. An always-on alarm detects every attack, so a detection rate without a false-positive rate is not a result. On normal GEM-CAN test traffic the proposed model raises false alarms on 2.26% of frames, against 0.03% to 0.31% for the baselines: about 340 false alarms per 15,063 normal frames. The target rate at calibration is 1%, and the realised rate is higher for two reasons: false alarms from the supervised head are counted as alarms of the system, and the normal traffic used for validation is not the normal traffic used for test.

Fig. 6 shows the whole trade-off rather than the single point. On CICIoV2024 the curve is effectively a step: detection reaches 100% at a false-alarm rate below the sweep resolution and stays there, with an area under the curve of 1.000 for all three quarantined classes, so the operating point reported in Table VIII is conservative and could be lowered by an order of magnitude at no cost in detection. The GEM-CAN run available to us recorded the operating point only

and not the full sweep, so the corresponding curve for that capture is left for future work rather than estimated here.

The operationally meaningful unit is alarms per unit time. At 2.26%, a bus carrying f frames per second produces 0.0226f alarms per second. At any frame rate a production powertrain bus sustains, that is more alarms than an operator or a downstream controller could triage. We therefore present the gate at this operating point as a research instrument rather than a deployable alarm. The threshold is the knob available, and a deployment moves down the detection-cost curve by lowering the target false-positive rate; what the tolerable rate is depends on a per-vehicle frame rate and a per-fleet triage budget that these captures do not supply.

The false negatives are concentrated and explicable rather than scattered. On GEM-CAN every missed frame of the quarantined class is zero, because the gate detects all 119; the systematic false negative is steering tampering, which the gate misses entirely for the reason given in Section VI-E. On CICIoV2024 there are no false negatives at all: every frame of every quarantined class raises an alarm, and the false-alarm rate falls to 0.37%.

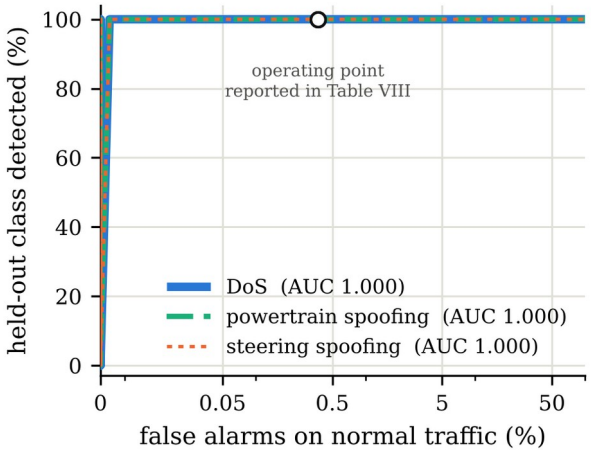


FIGURE 6. Detection of the quarantined class against false alarms on normal traffic, as the gate threshold is swept, on CICIoV2024. The three curves coincide and are drawn with different dash patterns for that reason; the area under each is 1.000. The circle marks the operating point reported in Table VIII, which is conservative: detection is already complete at a far smaller false-alarm rate.

G. ABLATION STUDY

Table V isolates the contribution of each component on the same CNN-BiLSTM backbone. The ladder is not monotonic with the identifier included, and we read it as it stands: macro-F1 falls from 0.9979 to 0.9958 when the scalar twin output is added, and only the full per-byte residual recovers and surpasses the backbone at 0.9989. With the identifier excluded the ladder does rise at each step, from 0.9914 to 0.9965 to 0.9979. In both variants adaptive fusion adds nothing beyond the residual, 0.9979 to 0.9979.

The conclusion we draw is narrower than the architecture would invite. In this experiment the gain lies in the residual, not in the attention layer, and we report it that way. The decomposition of Section VI-A reinforces it: on the quarantined class the attention-weighted fusion contributes nothing at all, because the detection comes entirely from the gate.

H. EXPLAINABILITY ANALYSIS

Every alarm raised by the gate carries three pieces of evidence without any additional model: which of the two views fired, which arbitration identifier carried the largest standardised residual, and which payload byte within it. On GEM-CAN brake tampering this localisation is unambiguous, with bytes DATA_4 and DATA_5 of identifier 18FF00F9 carrying almost the whole deviation, which corresponds to the forged brake command. An operator receiving such an alarm is told the ECU and the signal, not merely that something is wrong.

Shapley additive explanation values [27] computed over the supervised branch attribute its decisions across payload bytes, identifier and timing features, and confirm that with the identifier available the model leans heavily on it for the volumetric class, which is the quantitative form of the shortcut documented in Section III. With the identifier removed the attributions redistribute onto the payload bytes. Both tools identify where the evidence lies rather than why a payload is surprising; the latter would require the vehicle's signal database, as noted in Section IV-F.

I. COMPUTATIONAL EFFICIENCY

A forward pass through all four models plus one gate evaluation takes 12.0 ms per frame on CPU, approximately 3 ms per model. The gate itself is two standardisations and a comparison and is negligible beside the twin's forward pass. The digital twin adds one GRU step per frame over the supervised branch, which is the price of the open-set capability.

We do not claim line-rate operation on vehicle hardware. The system was not designed for an embedded target, the measurement is single-threaded CPU on a development machine, and a production deployment would need quantisation, batching across identifiers, or an accelerator. What the figure does establish is that the residual channel is not the bottleneck: the cost is dominated by the supervised backbones that any CAN detector of this family already pays.

VII. DISCUSSION

A. KEY FINDINGS

Three findings follow from the experiments. First, closed-set accuracy has stopped discriminating between CAN intrusion detectors on these benchmarks: four architectures agree to within 0.0031 macro-F1, and a lookup table on the arbitration identifier alone reaches 98.9% accuracy on GEM-CAN. Reporting accuracy as the headline on such a benchmark measures the dataset, not the model.

Second, the same four models separate by more than 74 percentage points when one attack class is quarantined from every training stage, and the separation is attributable to one component. The supervised head contributes none of the

detection of the unseen class, because it has no label to assign; the digital-twin residual gate contributes all of it. A channel fitted without labels generalises to an attack nobody labelled, and a channel fitted with labels does not.

Third, that generalisation has a precise and measurable boundary. The gate detects an attack if and only if the attack drives the payload outside the range the twin has learned for that identifier. Brake tampering on GEM-CAN raises the residual by a factor of 9.5 and is detected completely; steering tampering on the same capture lowers it and is missed completely. The condition is a property of the attack, not of the detector's capacity, and no additional model depth changes it.

B. CROSS-DATASET GENERALIZATION

The method transfers across both real vehicles in this study. On GEM-CAN it detects the quarantined brake-tampering class completely where the best baseline reaches 25.21%, and on CICIoV2024 it detects every one of three quarantined classes completely at a false-alarm rate of 0.37%. Two captures, two manufacturers, two attack families, one protocol, and the same outcome.

The useful conclusion from the cross-dataset work is therefore not a performance number but a predicate. Before deploying a residual gate on a new vehicle or a new capture, one can ask whether the attacks of concern place implausible values on the bus. If they do, the gate transfers. If they hold signals at plausible values, it does not, and no amount of retraining on normal traffic will change that, because the twin is already modelling normality correctly. This is a sharper and more portable statement than an accuracy figure on a second dataset.

C. PRACTICAL IMPLICATIONS FOR EV CYBERSECURITY

For an electric vehicle fleet, three implications follow. A residual gate of this kind is best understood as a second line rather than a replacement: it should run alongside a supervised classifier, which remains the precise and cheap detector for attacks that have been catalogued, and should be the component that catches what the catalogue misses. Deploying the gate alone would trade precision for coverage unnecessarily.

The false-alarm budget is the binding constraint, not accuracy. At a 2.26% rate on a bus carrying thousands of frames per second, the alarm stream exceeds any plausible triage capacity, so an operational deployment must either lower the target rate and accept reduced sensitivity, or aggregate alarms over a window before escalating. The per-identifier, per-byte localisation described in Section VI-H is what makes such aggregation tractable, because alarms can be grouped by the ECU and signal they implicate rather than counted in bulk.

Finally, the attacks that matter most on an electric, drive-by-wire platform are precisely the hard ones. A denial-of-service flood is disruptive and easy to detect; a forged steering or torque command is dangerous and, when its values are plausible, invisible to a payload residual. A deployment that must cover that case needs a timing or sequence model alongside the twin, because the evidence for such an attack is in when and how often a frame arrives rather than in what it contains.

D. LIMITATIONS AND THREATS TO VALIDITY

The method has two boundaries. Detection requires payload anomaly, as established in Section VI-E, and a replay attack would evade the gate for the same reason: replaying a genuine payload presents the twin with a predictable payload, so the residual stays low. Replay detection is timing or sequence modelling rather than payload modelling. A third case, an attack arriving on an arbitration identifier the twin has never observed, was untested in earlier work and is resolved here: every CICIoV2024 attack identifier is absent from the benign capture, and the gate detects all three such classes completely, because the twin holds no normal profile for them and the standardised residual is correspondingly large.

The evaluation has four limits of its own. The block-wise split leaves adjacent blocks temporally correlated, and 87.4% of GEM-CAN frames are byte-identical to another frame, with 80.6% of test frames having an exact twin in training; both inflate frame-level metrics equally for all models, so the leave-one-attack-out comparison is unaffected, but both should temper any reading of the absolute closed-set figures. All figures come from a single seeded run: the 74-point gap that carries the argument is far larger than any plausible seed variance, while the differences of a few thousandths in Tables IV and V are not, and we claim no ordering among the four architectures. The CICIoV2024 quarantine was repeated over four seeds, which bounds that concern there but not on GEM-CAN, where a repeat would require access to the raw capture. The baseline set contains three supervised architectures and does not include the unsupervised comparisons a reader might reasonably expect: a plain autoencoder, a one-class model, an explicit open-set scoring rule, or a per-identifier range check, which is the simplest thing that could detect brake tampering on this capture. And validation covers two real captures plus one synthetic benchmark; Car-Hacking, OTIDS and ROAD remain untested and would each require the same conversion and the same four-stage quarantine before they could be reported.

The most serious threat to validity is the redundancy of the benchmarks themselves, quantified in Section II-C. When 99.9% of a capture's frames are byte-identical to another frame and the whole release holds 3,588 distinct identifier-payload tuples, a frame-level score close to 1.000 is substantially a measurement of the benchmark. This affects every detector evaluated on these captures, ours included, and it is the reason we lead with the leave-one-attack-out alarm rate rather than with closed-set accuracy, and the reason we treat GEM-CAN rather than CICIoV2024 as the load-bearing evaluation. A capture with per-frame labels, recorded timestamps and substantially more distinct

payloads would be needed to resolve detection capability beyond the level these benchmarks can support.

A further threat is specific to CICIoV2024 and is restated here so that it cannot be overlooked: the release carries no timestamp, so the inter-arrival feature is synthesised and constant on that dataset. Any conclusion that depends on timing is therefore not supported by the CICIoV2024 results, and the comparison in Section VI-C is read as a comparison of payload-based detection only.

VIII. CONCLUSION AND FUTURE WORK

Closed-set evaluation has told the CAN intrusion detection literature that the problem is nearly solved. The four architectures compared here sit within 0.0031 macro-F1 of one another on the dataset studied, which says more about the benchmark than about the models, and a lookup table on the arbitration identifier alone reaches 98.9% accuracy on the same capture.

The same four models separate by more than 74 percentage points when one attack class is quarantined from the label space, from the digital twin's history and from the anomaly threshold's calibration set. The digital-twin residual gate, calibrated on normal traffic alone, raises an alarm on every held-out frame of the quarantined class at a 2.26% false-positive rate, where the best supervised baseline reaches 25.21%. The mechanism has a well-defined boundary and we have measured it rather than asserted it: detection rests on the attack being anomalous in the payload, and an attack whose forged values fall within the learnt-normal range is not detected. Evaluation on a second real capture, CICIoV2024 from a 2019 Ford, and on the synthetic SynCAN benchmark, tests that boundary rather than merely repeating the headline figure.

Future work follows the limitations directly. The evaluation should be extended across vehicles and drive cycles, and across the remaining public captures, so that the transfer predicate of Section VII-B can be tested rather than inferred. The baseline set should be widened to include a per-identifier range check, an autoencoder and an explicit open-set scoring rule, so that the residual gate is measured against the simplest alternatives as well as against supervised ones. The false-alarm rate needs to come down by an order of magnitude before the gate is deployable, which points at alarm aggregation over a window and at a better residual representation. And the attacks the residual cannot see, those that mimic normal payload statistics, require a timing or sequence channel alongside the twin; that is where the remaining headroom lies.

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